

OSH

PHYSICAL CROSS-ARCHITECTURE EVIDENCE

From reference execution to physical embedded silicon

PC → UNIVERSAL SUBSTRATE → ESP32-S3 → ARM CORTEX-M4

Public technical whitepaper. Observable behavior, physical targets, test methodology and measured results are disclosed. The proprietary OSH kernel, internal algorithms, equations and source-level black-box implementation are not.

Organization	OSH Global Systems, Inc.
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Evidence focus	Portability, physical execution, parity, real I/O, timing and board-level power
Disclosure boundary	Evidence public; proprietary mechanism private

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Abstract

OSH is an embedded AI architecture being developed for deterministic, local operation without dependence on cloud inference, external APIs or an LLM runtime. This whitepaper addresses a specific engineering question: can the expected behavior of a frozen computational substrate be preserved as execution moves across materially different architectures and then onto physical embedded hardware?

The evidence chain progresses from PC reference work and a Universal Substrate Benchmark to physical ESP32-S3 execution and physical STM32F446RE ARM Cortex-M4 execution. The ARM milestone adds native execution, a 30-case parity gate, a 100,000-cycle physical DAC-to-ADC closed loop, external oscilloscope observation, artifact identity and board-level input-power observation.

Compilation is not treated as execution. Board-level power is not presented as isolated OSH-core power. Oscilloscope cursor coincidence is not presented as zero physical latency.

Evidence chain

Stage	Evidence	Status
PC / Benchmark 3	Reference execution and measured baseline	Demonstrated
Universal Substrate Benchmark	Frozen Tier-A substrate across multiple ISA profiles; RV32IMF golden-case demonstration	Demonstrated
ESP32-S3	First physical MCU migration and embedded HAL validation	Demonstrated
STM32F446RE	Physical ARM Cortex-M4 execution, parity, 100K loop and external instrumentation	Demonstrated
Physical RISC-V	Next physical cross-architecture milestone	Next

1. Why physical cross-architecture validation matters

Portability has distinct levels. Source compatibility shows that a toolchain can accept code. Object generation shows that target instructions can be emitted. Emulator or harness execution shows behavior under a model. Physical execution adds the real instruction set, floating-point behavior, memory organization, clocks, peripherals, electrical interfaces and runtime constraints of actual silicon.

OSH therefore treats 'compiled for another architecture' and 'verified on another physical architecture' as different claims.

Level	Demonstrates	Does not by itself demonstrate
Compile	Target toolchain accepts source	Correct runtime behavior
Link / image	Loadable target artifact exists	Physical execution
Emulated execution	Behavior under modeled target	Electrical / peripheral behavior
Physical execution	Native runtime on real silicon	Universal correctness
Physical parity + instrumentation	Observed behavior plus real I/O evidence	Independent certification

2. From PC to portable substrate

Benchmark 3 established an early PC evidence baseline. The subsequent Universal Substrate Benchmark separated architecture portability from a single development machine: a frozen Tier-A substrate generated objects for 21 of 22 tested target profiles spanning x86, ARM, RISC-V RV32IMF, MIPS, PowerPC, SPARC, LoongArch64 and WASM32.

A Tier-A harness executed under RV32IMF and reproduced both frozen golden cases with a recorded maximum absolute error no greater than 5×10^{-8} . This stage was evidence of portable substrate behavior; it was not represented as physical execution on every listed architecture.

That distinction is what motivated the next step: physical MCU migrations.

3. First physical migration - ESP32-S3

The first physical embedded migration moved OSH from PC/reference execution into a real ESP32-S3 microcontroller environment. The work established a repeatable method: freeze the evidence boundary, port the required runtime components, execute on physical silicon, verify the expected behavior and preserve reproducible evidence without exposing the proprietary kernel.

Physical reproduction was achieved on two ESP32-S3 devices, with embedded validation including the Tier-B1 HAL boundary and scaling work reaching 1,024 candidates. This milestone demonstrated that OSH could leave the PC environment and operate within MCU-class constraints.

The significance of ESP32-S3 was therefore not simply support for one board. It was the first physical proof that the architecture could cross the PC-to-microcontroller boundary.

4. Second physical migration - STM32F446RE / ARM Cortex-M4

The second physical migration placed OSH on an ST NUCLEO-F446RE carrying an STM32F446xx ARM Cortex-M4 MCU. This is a materially different embedded platform and therefore strengthens the cross-architecture evidence chain.

ARM validation	Measured result
Target	NUCLEO-F446RE / STM32F446xx / ARM Cortex-M4
Programmer verification	Download verified successfully
100K image size	67.89 KB
Structural parity	30 / 30 PASS
Physical loop	DAC A2/PA4 → external electrical path → ADC A0/PA0
Physical cycles	100,000
ADC failures / state failures	0 / 0
Actions	INC 36 / HOLD 99,934 / DEC 30
Elapsed time	24.937 s
Overall ARM 100K	PASS

This is physical ARM execution evidence, not a cross-compile-only or emulator result.

5. Physical ARM closed-loop results

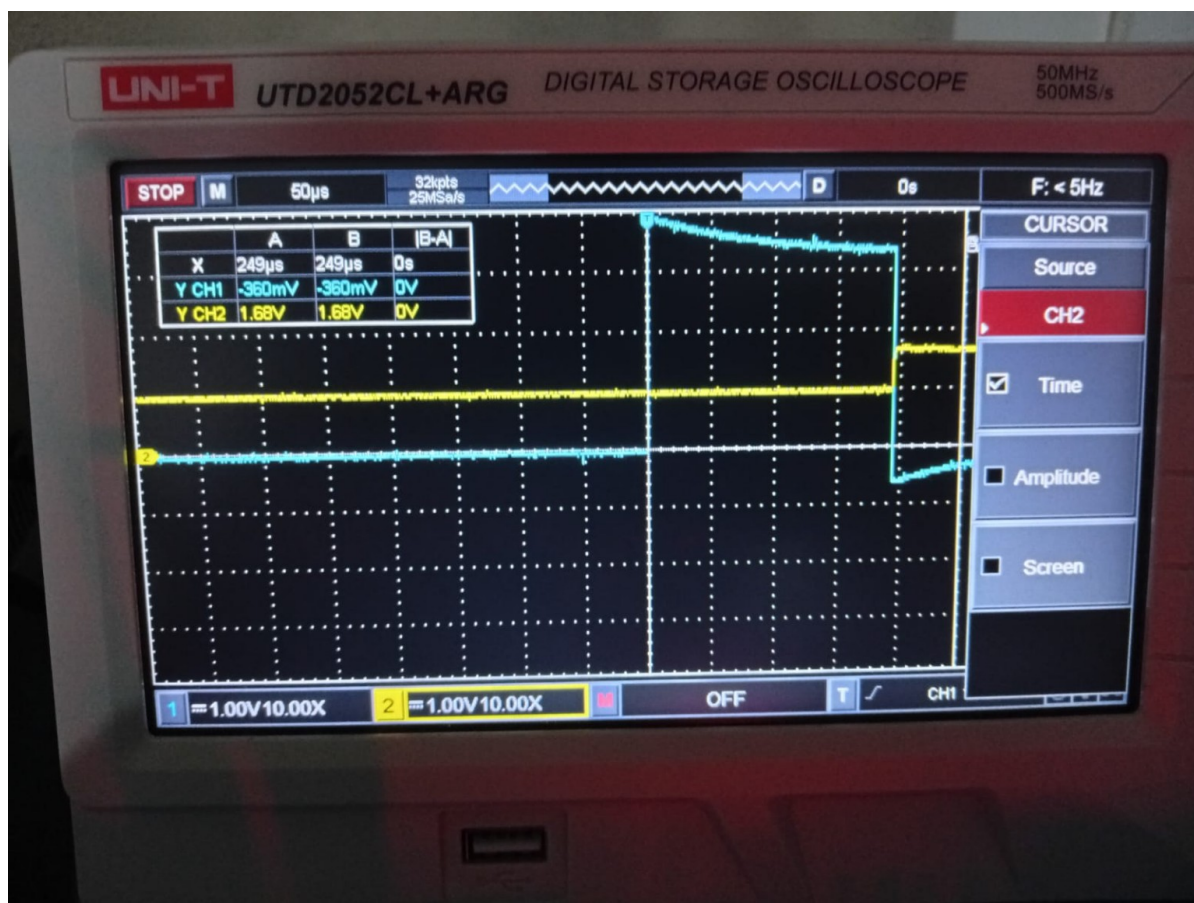
Before the long run, the target passed a three-level electrical loop probe. It then completed the 100,000-cycle DAC-to-ADC closed loop with zero reported ADC failures and zero state failures.

Measurement	Result
Electrical low / mid / high ADC	512 / 2053 / 3590
Electrical loop gate	PASS
ADC range	1243 - 2759
DAC range	1280 - 2720
Maximum DAC-ADC mismatch	43 counts
Final ADC / DAC / reference	2343 / 2336 / 2662
Final error	319 counts
Maximum state delta	106815 ppm
Physical 100K gate	PASS

A real analog output left the MCU, crossed an external electrical connection and returned through the ADC. The test is therefore described as a physical electrical closed loop, not a simulation-only loop.

6. External timing evidence

The ARM firmware measured the instrumented route with the Cortex-M cycle counter: 209.500 μs minimum, 241.875 μs maximum and 241.250 μs mean. A separate oscilloscope experiment externally observed the instrumented execution window.

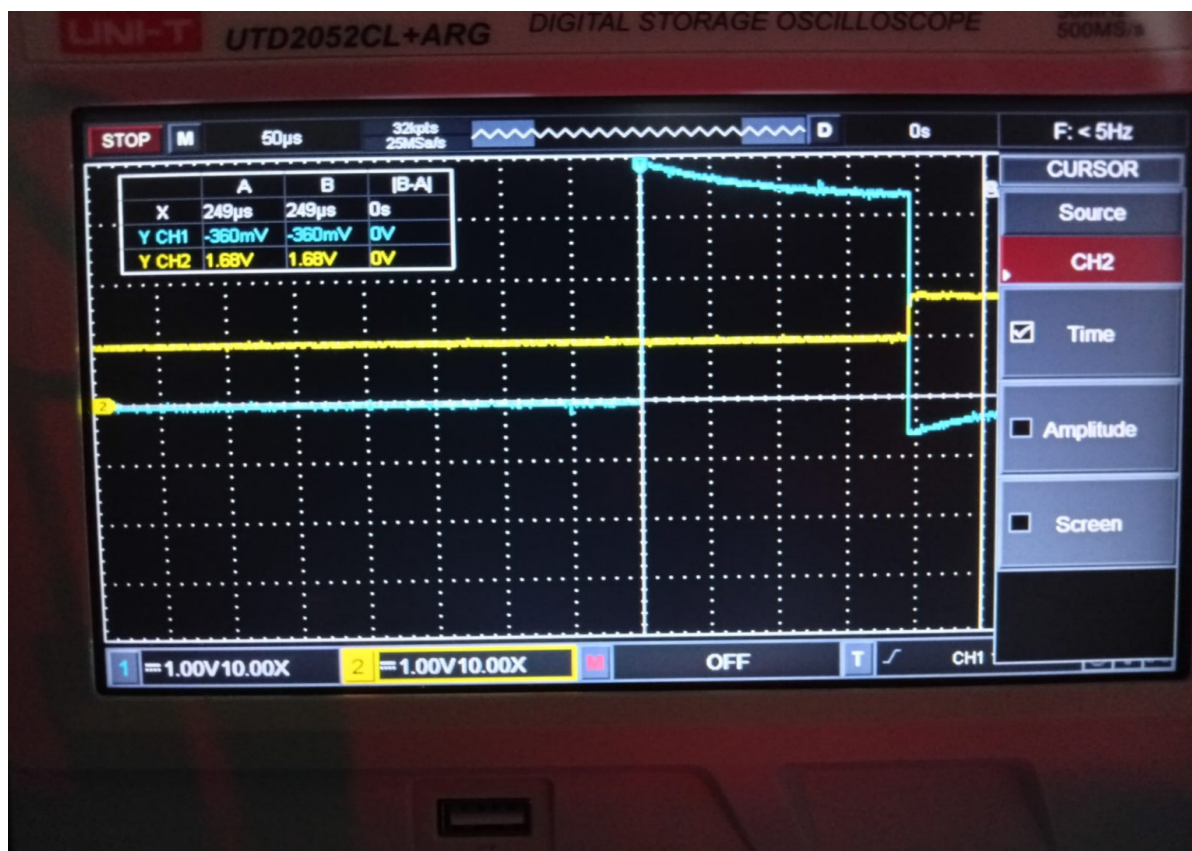


External oscilloscope capture. CH1 (cyan) marks the instrumented execution window; CH2 (yellow) observes the analog-output event.

The externally measured CH1 pulse width was **226 μs** at 50 $\mu\text{s}/\text{div}$. It is reported as an independent external measurement on the same timing scale, not as a claim that the oscilloscope and internal cycle-counter boundaries are perfectly identical.

7. External causal-path observation

A dual-channel instrumentation experiment observed the execution marker and analog output simultaneously. For visibility, the DAC step in this specific scope experiment was deliberately enlarged; its amplitude is not the normal 100K control step.



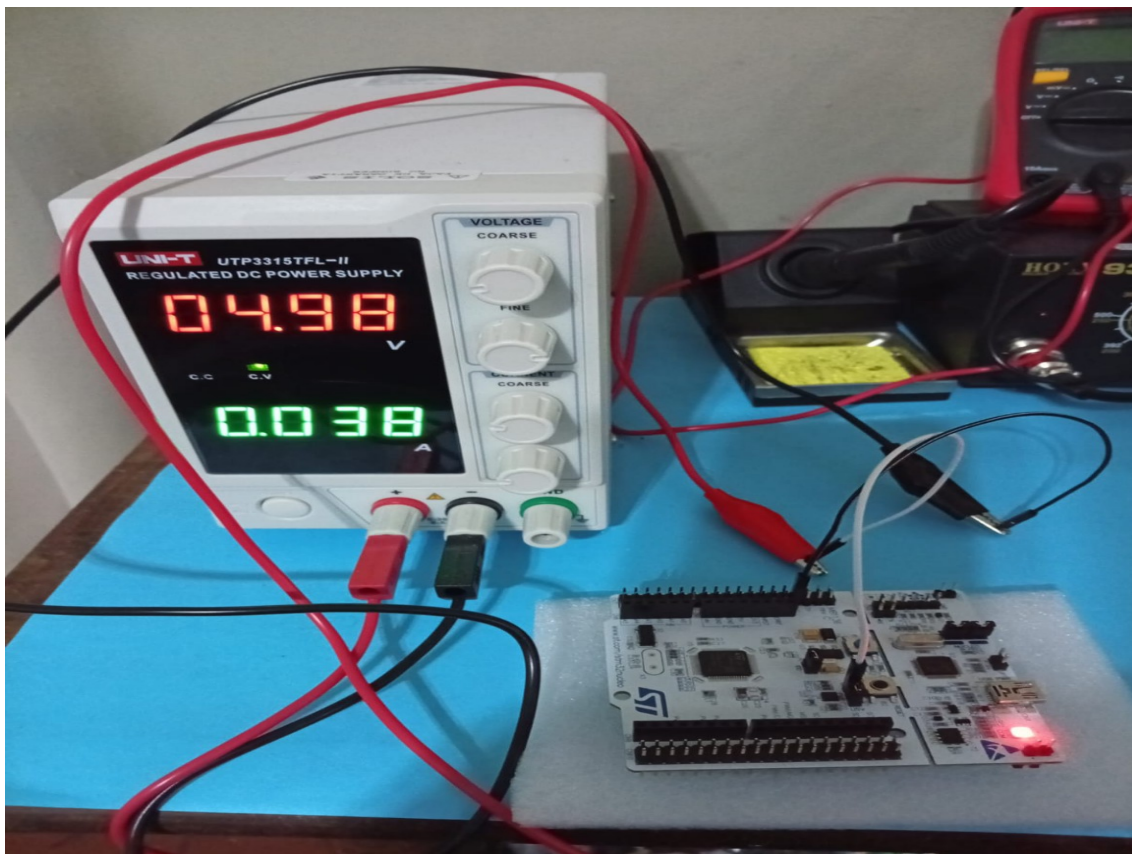
Dual-channel frozen trace: execution marker and analog-output transition observed externally.

The final cursor capture showed $A = 249 \mu\text{s}$, $B = 249 \mu\text{s}$ and $|B-A| = 0 \text{ s}$. The correct interpretation is coincident within the time resolution of the captured trace; this is not a claim of zero physical latency.

8. Board-level input power - supporting measurement

Power is a supporting measurement, not the central claim of this whitepaper. After restoring and verifying the 100K firmware, USB power was removed and the NUCLEO was powered exclusively from an external regulated UNI-T supply at 4.98 V.

During the run window the supply display was observed varying approximately from 27 mA to 38 mA, corresponding to approximately 134.5 mW to 189.2 mW at the board input.



Supporting energy evidence. The photographed instant shows the external supply at 4.98 V and 0.038 A during the physical ARM validation.

A ~28.2 s video covered essentially the complete ~24.94 s 100K execution window. These figures are board-level input observations, including the NUCLEO platform electronics. They are not isolated OSH-core power and are not an energy-efficiency certification.

9. Artifact integrity and disclosure boundary

The validation program preserves hashes and frozen evidence so later reviewers can establish artifact identity without receiving the proprietary black-box mechanism.

Public evidence	Remains proprietary
Target hardware and toolchain identity	OSH kernel implementation
Test topology and external electrical path	Internal algorithms and equations
Parity, gate, timing and failure results	Semantic-state construction mechanisms
Oscilloscope observations	Decision internals and proprietary selectors
Board-level input-power observation	Source-level black-box architecture
Artifact hashes	Confidential production code

The purpose of the public evidence package is to make externally testable claims visible while preserving the proprietary mechanism that produces them.

10. Conclusion

The ARM milestone changes the status of OSH portability evidence. ARM is no longer represented only as a compilation target: a physical STM32F446RE executed the validation path, passed the 30-case parity gate, operated a real DAC-to-ADC electrical loop for 100,000 cycles with zero reported ADC or state failures, and produced externally observable timing and output events.

Taken together, the evidence now forms a progression from PC reference execution, through a portable substrate benchmark, to physical ESP32-S3 and physical ARM Cortex-M4 execution.

This does not establish universal hardware portability or third-party certification. The next high-value milestone is physical RISC-V execution under the same frozen validation discipline.

OSH Global Systems, Inc.

Public technical document - proprietary implementation details intentionally omitted.